



LINUX FUNDAMENTALS & CLI BASICS

1. WHAT IS LINUX?

- **Family** of open-source, Unix-like operating systems built on the **Linux kernel**
- An **Operating System (OS)** acts as the interface between user and hardware components

2. CLI = COMMAND-LINE INTERFACE

- **Text-based interface** where you type commands → computer executes them
- **Essential for DevOps** for speed, automation, and control

3. SEVEN COMMAND CATEGORIES

Category	Purpose	Examples
System	System management & control	reboot , shutdown , systemctl
Hardware	Hardware information & monitoring	lscpu , lsblk , free , df
File	File and directory operations	ls , cp , mv , rm , mkdir
Permission	Access control & security	chmod , chown , umask
User	User and group management	useradd , usermod , groupadd
Search	Finding files and content	grep , find , locate , which
Software	Package management	apt , yum , dnf , pip

🧠 Memory Hook: S-H-F-P-U-S-S

- System → Hardware → File → Permission → User → Search → Software

4. ESSENTIAL SHORTCUTS & PRIVILEGE MANAGEMENT

Privilege Escalation

Task	Command
Become root user	sudo -i or sudo su -

Task	Command
Run single command as root	<code>sudo <command></code>
Return to normal user	<code>exit</code>
Switch to another user	<code>su - username</code>

Terminal Shortcuts

Shortcut	Action
<code>Ctrl + L</code> or <code>clear</code>	Clear terminal screen
<code>Ctrl + C</code>	Terminate current process
<code>Ctrl + D</code>	Logout/exit shell
<code>Ctrl + A</code>	Move to beginning of line
<code>Ctrl + E</code>	Move to end of line
<code>Ctrl + U</code>	Clear line before cursor
<code>Ctrl + K</code>	Clear line after cursor
<code>Tab</code>	Auto-complete commands/files

5. BASIC COMMAND STRUCTURE

```
command [options] [arguments]
```

- **Command:** The program to execute
- **Options:** Modify behavior (usually `-` or `--`)
- **Arguments:** Targets or parameters for the command

6. QUICK START GUIDE

Immediate Practice Commands

```
# Check current user
whoami

# See current directory
pwd

# List files
```

```
ls

# Check system information
uname -a

# View disk usage
df -h

# Check memory
free -h
```

Navigation Basics

```
# Change directory
cd /path/to/directory

# Go to home directory
cd ~ OR cd

# Go up one level
cd ..

# Previous directory
cd -
```

7. STUDY TIPS

For Beginners:

1. **Practice daily** - Use the terminal for common tasks
2. **Learn help system** - Use `man command` or `command --help`
3. **Start with file operations** - Master `ls`, `cd`, `cp`, `mv`, `rm`
4. **Understand permissions early** - Critical for security

For DevOps:

1. **Automate everything** - Script repetitive tasks
2. **Master text processing** - `grep`, `awk`, `sed`
3. **Learn process management** - `ps`, `top`, `kill`
4. **Practice privilege escalation** - Safe `sudo` usage



LINUX SYSTEM COMMANDS - COMPREHENSIVE CHEAT SHEET

1. OS & KERNEL INFORMATION

Task	Command	Sample Output
OS Name	<code>uname</code>	Linux
Kernel Version	<code>uname -r</code>	6.5.0-28-generic
System Architecture	<code>uname -m</code>	x86_64
Full System Details	<code>uname -a</code>	Linux host 6.5... x86_64 GNU/Linux

2. HOSTNAME MANAGEMENT

Action	Command
Show Current Hostname	<code>hostname</code>
Change Hostname	<code>sudo hostnamectl set-hostname "my-server"</code>
Note: Logout → login to see changes	

3. NETWORK CONFIGURATION

Information	Command
All Network Addresses	<code>ip addr</code> or <code>ip a</code>
Routing Table	<code>ip route</code>
Network Interfaces	<code>ip link show</code>
Legacy Tool (if installed)	<code>ifconfig</code>

4. CALENDAR & DATE COMMANDS

Calendar Views		
Task	Command	Description
Current Month	<code>cal</code>	Grid calendar for current month
Specific Year	<code>cal 2024</code>	Entire year calendar
Specific Month/Year	<code>cal 8 2012</code>	August 2012 calendar

Date Formatting

Format	Command	Example Output
Day of Month	<code>date +%d'</code>	05
Month Number	<code>date +%m'</code>	12
Year (4-digit)	<code>date +%Y'</code>	2024
Hours (24h)	<code>date +%H'</code>	19
Minutes	<code>date +%M'</code>	34
Seconds	<code>date +%S'</code>	07
Full Date (Slash)	<code>date +%d/%m/%y'</code>	12/05/24
ISO-8601 Format	<code>date +%F'</code>	2024-12-05
Day Name	<code>date +%A'</code>	Thursday
Month Name	<code>date +%B'</code>	December
Custom Format	<code>date +%A, %B %d, %Y'</code>	Thursday, December 05, 2024

5. TIME & TIMEZONE MANAGEMENT

Action	Command
Show Current Time + Zone	<code>timedatectl</code>
List Available Timezones	<code>timedatectl list-timezones</code>
Change Timezone	<code>sudo timedatectl set-timezone Asia/Kolkata</code>
Set Hardware Clock	<code>sudo timedatectl set-local-rtc 1</code>

6. USERS & PROCESSES

Need	Command
Current Username	<code>whoami</code>
All Logged-in Users	<code>who</code> or <code>w</code>
Running Processes (Simple)	<code>ps</code>

Need	Command
All Processes	<code>ps aux</code>
Background Jobs	<code>jobs</code>
Kill Process	<code>kill -9 4430</code>
Kill by Name	<code>pkill process-name</code>

7. SYSTEM UPTIME & LOAD

Information	Command	Sample Output
Uptime with Load	<code>uptime</code>	14:20 up 5 days, load 0.20
Boot Time	<code>uptime -s</code>	2024-05-01 09:16:40
Pretty Format	<code>uptime -p</code>	up 5 days, 3 hours
System Load	<code>cat /proc/loadavg</code>	0.20 0.18 0.15 1/500 3412

8. PRIVILEGE & TERMINAL MANAGEMENT

Privilege Escalation

Task	Command
Become Root	<code>sudo -i</code> or <code>sudo su -</code>
Run Single Command as Root	<code>sudo command</code>
Return to Normal User	<code>exit</code>

Terminal Shortcuts

Shortcut	Action
<code>Ctrl + L</code> or <code>clear</code>	Clear Screen
<code>Ctrl + C</code>	Cancel/Interrupt
<code>Ctrl + D</code>	Exit Shell
<code>Ctrl + A</code>	Move to Line Start

Shortcut	Action
Ctrl + E	Move to Line End
Ctrl + U	Clear to Line Start
Ctrl + K	Clear to Line End
Tab	Auto-complete

9. QUICK SYSTEM CHECKS

One-Line System Info

```
# Comprehensive system overview
uname -a && hostname && date && uptime

# Memory and CPU info
free -h && lscpu | grep "Model name"

# Disk space check
df -h | grep -v tmpfs
```

10. PRACTICAL USAGE SCENARIOS

For System Administration:

- Use `timedatectl` for timezone management in cloud deployments
- Monitor system load with `uptime` during performance checks
- Use `ip addr` for network troubleshooting

For Scripting:

```
# Use date in backup scripts
backup_file="backup_$(date +%Y%m%d_%H%M%S').tar.gz"

# Check system uptime in monitoring scripts
if uptime -p | grep -q "day"; then
    echo "System has been up for multiple days"
fi
```

1. HARDWARE & RESOURCE INSPECTION

Resource	Command	Purpose & Output
CPU Information	<code>lscpu</code>	CPU model, cores, architecture, flags
Live System Monitor	<code>top</code>	Real-time CPU/MEM usage (press <code>q</code> to quit)
Memory Summary	<code>free</code>	RAM in KB
Memory in MB	<code>free -m</code>	RAM in megabytes
Memory in GB	<code>free -g</code>	RAM in gigabytes
Disk Space	<code>df -h</code>	Human-readable disk usage
Block Devices	<code>lsblk</code>	List storage devices (EBS, partitions, disks)
Hardware Info	<code>lshw</code>	Detailed hardware configuration

2. FILE TYPE LEGEND (`ls -l` FIRST CHARACTER)

Symbol	File Type	Examples
<code>-</code>	Regular file	Documents, scripts, images
<code>d</code>	Directory	Folders
<code>b</code>	Block device	Hard drives, EBS volumes
<code>c</code>	Character device	Terminals, serial ports
<code>l</code>	Symbolic link	Shortcuts to files/directories
<code>p</code>	Named pipe	Inter-process communication
<code>s</code>	Socket	Network communication

3. CREATING FILES & DIRECTORIES

File Creation

Need	Command	Result
Single file	<code>touch file1</code>	Creates empty file1
Multiple files	<code>touch file1 file2 file3</code>	Creates three files

Need	Command	Result
Sequential files	<code>touch file{1..7}</code>	Creates file1 through file7
Pattern files	<code>touch app_{1..3}.txt</code>	Creates app_1.txt, app_2.txt, app_3.txt

Directory Creation

Need	Command	Result
Single directory	<code>mkdir aws</code>	Creates aws folder
Multiple directories	<code>mkdir dir1 dir2 dir3</code>	Creates three directories
Sequential directories	<code>mkdir aws{1..7}</code>	Creates aws1 through aws7
Nested directories	<code>mkdir -p parent/child/grandchild</code>	Creates full path

4. LISTING ITEMS

Task	Command	Output
Simple list	<code>ls</code>	Basic file/directory names
Long format	<code>ls -l</code>	Detailed list with permissions
All files	<code>ls -a</code>	Includes hidden files (starting with .)
Human readable	<code>ls -lh</code>	File sizes in KB, MB, GB
Reverse order	<code>ls -lr</code>	Reverse alphabetical order
Sort by time	<code>ls -lt</code>	Newest files first
Recursive	<code>ls -R</code>	List subdirectories recursively
Common alias	<code>ll</code>	Typically <code>ls -l</code>

5. REMOVING FILES & DIRECTORIES

File Removal

Scope	Command	Behavior
Single file	<code>rm file1</code>	Asks for confirmation

Scope	Command	Behavior
Multiple files	<code>rm file1 file2 file3</code>	Asks for each file
Force delete	<code>rm -f file1</code>	No prompts, force removal
Recursive force	<code>rm -rf dir1</code>	Remove directory and contents
Range deletion	<code>rm -rf aws{1..7}</code>	Removes aws1 through aws7
All files	<code>rm -rf *</code>	Dangerous: Removes everything in current directory
Pattern deletion	<code>rm -rf a*</code>	Removes all files starting with "a"
Extension deletion	<code>rm -rf *.java</code>	Removes all .java files

Directory Removal

Situation	Command	Limitations
Empty directory	<code>rmdir dir1</code>	Only works on empty directories
Multiple empty dirs	<code>rmdir dir1 dir2 dir3</code>	Removes multiple empty directories
Note: <code>rmdir</code> provides no confirmation prompts		

6. SAFETY PRECAUTIONS ⚠️

Dangerous Commands to Avoid

```
# NEVER RUN THESE UNLESS ABSOLUTELY SURE
rm -rf /           # Deletes entire system
rm -rf *           # Deletes all files in current directory
rm -rf .*          # Deletes all hidden files
```

Safe Alternatives

```
# Always check before deleting
ls before_removing/
rm -ri directory/ # Interactive deletion
```

7. MEMORY HOOKS & TIPS

Command Associations

- `lsblk` = "List Block Devices"
- `lscpu` = "List CPU Information"
- `mkdir -p` = "Make directory with Parents"

Brace Expansion Magic

```
# Create multiple resources quickly
touch document_{a..c}.txt
mkdir server_{01..10}
rm -rf temp_{1..5}
```

Practical Scenarios

```
# Quick workspace setup
mkdir -p project/{src,bin,doc,test}
touch project/src/main_{1..3}.java

# Cleanup specific patterns
rm -rf *.log
rm -rf temp_*
```

8. QUICK REFERENCE CARD

Essential Hardware Commands

```
lscpu      # CPU info
free -h    # Memory in human format
df -h      # Disk space
lsblk      # Block devices
```

Essential File Operations

```
touch new_file.txt      # Create file
mkdir new_directory     # Create directory
ls -la                  # List all with details
rm -rf old_directory    # Remove directory
rm *.tmp                 # Remove all .tmp files
```



DIRECTORY NAVIGATION, FILE OPERATIONS & EDITORS

1. DIRECTORY NAVIGATION

Action	Command	Description
Show Current Directory	<code>pwd</code>	Print Working Directory
Change to Folder	<code>cd folder1</code>	Enter specific directory
Go to Root Directory	<code>cd /</code>	Navigate to system root
Return to Previous Directory	<code>cd -</code>	Switch between current and last directory
Move Up One Level	<code>cd ..</code>	Go to parent directory
Move Up Two Levels	<code>cd ../../</code>	Go to grandparent directory
Navigate Deep Path	<code>cd folder1/folder2/folder3</code>	Direct multi-level navigation
Go to Home Directory	<code>cd ~</code> or <code>cd</code>	Return to user home directory

2. FILE & DIRECTORY CREATION

Quick Creation Commands

Need	Command	Result
Create File in Specific Path	<code>touch folder/deep.txt</code>	Creates file in target folder from anywhere
Preview Folder Contents	<code>ll folder</code>	Lists directory contents before entering
Create Nested Directories	<code>mkdir -p folder1/folder2/folder3</code>	Creates full path with parent directories
Key Option: <code>-p</code> = "create parents if absent"		

Practical Examples

```
# Create project structure
mkdir -p myproject/{src,bin,doc,test,logs}

# Create multiple config files
touch myproject/config/{app,db,web}.config

# Verify structure
ll myproject/
```

3. REMOVAL OPERATIONS ⚠

Dangerous Deletion Commands

Target	Command	Effect & Warning
Everything in Current Directory	<code>rm -rf *</code>	DANGEROUS: Removes all files and directories recursively
All Subdirectories Only	<code>rm -rf */</code>	Removes all directories but leaves files
Current Directory Contents	<code>rm -rf .</code>	EXTREME DANGER: Removes everything including hidden files

Safety First Practices

```
# Always check before deleting
ls before_removing/

# Use interactive mode for safety
rm -ri directory/

# Test with dry run first (if available)
rsync -av --delete --dry-run source/ empty_dir/
```

4. COPY & MOVE OPERATIONS

Copy Commands

Scenario	Command	Usage
Copy Single File	<code>cp src.txt dest.txt</code>	Basic file copy
Copy Multiple Files to Directory	<code>cp file1 file2 file3 dir/</code>	Last argument must be destination directory
Recursive Copy	<code>cp -r source_dir/ dest_dir/</code>	Copy directories and contents
Preserve Attributes	<code>cp -a source/ dest/</code>	Copy with permissions and timestamps

Move/Rename Commands

Scenario	Command	Usage
Move File	<code>mv old.txt new.txt</code>	Rename or relocate file
Move Multiple Files	<code>mv file1 file2 file3 dir/</code>	Move several files to directory
Bulk Renaming	<code>mv *.txt text_files/</code>	Move all .txt files to directory

Practical Examples

```
# Backup configuration
cp -a /etc/myapp/ /backup/myapp_backup/

# Organize files
mkdir -p images/ documents/
mv *.jpg *.png images/
mv *.pdf *.doc documents/

# Rename with pattern
mv old_filename.txt new_filename.txt
```

5. CAT COMMAND TRICKS 🐱

Action	Command	Usage
View File Contents	<code>cat file1</code>	Display entire file
Create/Overwrite File	<code>cat > file1</code>	Type content, press <code>Ctrl+D</code> to save
Append to File	<code>cat >> file1</code>	Add content, press <code>Ctrl+D</code> to save
View Multiple Files	<code>cat file1 file2</code>	Concatenate and display
Number Lines	<code>cat -n file1</code>	Show with line numbers
Display Non-printable	<code>cat -A file1</code>	Show all characters including endings

Advanced Cat Usage

```
# Create multi-line file
cat > script.sh << EOF
#!/bin/bash
echo "Hello World"
date
EOF

# Combine files
cat file1.txt file2.txt > combined.txt

# View with pagination
cat long_file.txt | less
```

6. VIM EDITOR - ESSENTIALS

Basic Vim Operations

Task	Command/Key	Description
Open File	<code>vim filename</code>	Start editing file
Modes	Command Mode (default)	For navigation, deletion, search
Save & Exit	<code>:wq</code> or <code>:x</code>	Write and quit
Save Only	<code>:w</code>	Write without exiting
Quit Without Save	<code>:q!</code>	Force quit ignoring changes
Navigation	<code>G</code>	Jump to last line
	<code>gg</code>	Jump to first line
	<code>:5</code>	Jump to line 5
Insert Mode	<code>i</code>	Insert before cursor
	<code>a</code>	Append after cursor
	<code>o</code>	Open new line below

Essential Vim Commands

```
# Basic editing workflow
vim file.txt      # Open file
i                # Enter insert mode
[type your content] # Edit text
Esc              # Return to command mode
:wq              # Save and exit

# Quick navigation
20G              # Go to line 20
/text            # Search for "text"
n                # Next search result
```

7. MEMORY HOOKS & BEST PRACTICES

Quick Mnemonics

- `cd -` = "Dash back to previous directory"
- `mkdir -p` = "Make parent directories"
- `rm -rf` = "Recursive Force - use with extreme caution"
- `cat > file` = "Create file like a cat scratching content"

Safety Protocols

```
# Always verify paths
pwd && ls -la

# Use aliases for safety
alias rm='rm -i'      # Interactive removal
alias cp='cp -i'      # Interactive copy

# Backup before major operations
cp important_file.txt important_file.txt.backup
```

Productivity Tips

```
# Quick directory navigation
cd /var/log && pwd

# Batch operations with brace expansion
touch report_{jan,feb,mar}.txt
mkdir -p project/{2023,2024}/{q1,q2,q3,q4}

# Use history for complex commands
history | grep "cp -r"
```

 VIM POWER MOVES - COMPREHENSIVE CHEAT SHEET

1. FAST NAVIGATION (COMMAND MODE)

Vertical Navigation

Jump To	Key(s)	Description
Top of File	<code>gg</code>	First line of file
Bottom of File	<code>G</code>	Last line of file
Middle of File	<code>M</code>	Middle of visible screen
Specific Line	<code>7gg</code> or <code>7G</code>	Jump to line 7
Specific Line (Command)	<code>:18</code> + <code>Enter</code>	Jump to line 18
Relative Line	<code>10j</code>	Jump 10 lines down
Relative Line	<code>5k</code>	Jump 5 lines up

Horizontal Navigation

Movement	Key(s)	Description
Start of Line	<code>0</code>	First character of line
First Non-blank	<code>^</code>	First non-whitespace character
End of Line	<code>\$</code>	Last character of line
Word Forward	<code>w</code>	Next word start
Word Backward	<code>b</code>	Previous word start
End of Word	<code>e</code>	End of current word

Screen Navigation

Action	Key(s)	Description
Half Page Down	<code>Ctrl + d</code>	Scroll down half page
Half Page Up	<code>Ctrl + u</code>	Scroll up half page
Full Page Down	<code>Ctrl + f</code>	Scroll down full page
Full Page Up	<code>Ctrl + b</code>	Scroll up full page
Center Screen	<code>zz</code>	Center current line
Toggle Line Numbers	<code>:set number</code>	Show/hide line numbers
Toggle Relative Numbers	<code>:set relativenumber</code>	Show relative line numbers

2. YANK / PASTE / DELETE OPERATIONS

Copy (Yank) Operations

Action	Key(s)	Result
Copy Current Line	<code>yy</code>	Yank entire line
Copy 5 Lines	<code>5yy</code>	Yank next 5 lines
Copy to End of Line	<code>y\$</code>	Yank from cursor to line end
Copy Word	<code>yw</code>	Yank current word
Copy to Character	<code>ytx</code>	Yank until character 'x'

Paste Operations

Action	Key(s)	Result
Paste After Cursor	p	Paste after current position
Paste Before Cursor	P	Paste before current position
Paste 5 Times	5p	Paste clipboard 5 times
Paste from System Clipboard	" + p	Paste from system clipboard

Delete Operations

Action	Key(s)	Result
Delete Current Line	dd	Cut entire line
Delete 10 Lines	10dd	Cut next 10 lines
Delete to End of Line	d\$	Cut from cursor to line end
Delete Word	dw	Cut current word
Delete Character	x	Delete character under cursor

Undo/Redo

Action	Key(s)	Description
Undo	u	Undo last change
Redo	Ctrl + r	Redo undone change
Undo Line	U	Undo all changes on current line

3. INSERT MODE SHORTCUTS

Entering Insert Mode

Action	Key(s)	Description
Insert at Cursor	i	Insert before cursor
Append After Cursor	a	Insert after cursor
Insert at Line Start	I	Insert at beginning of line

Action	Key(s)	Description
Append at Line End	A	Insert at end of line
New Line Below	o	Open new line below current
New Line Above	O	Open new line above current
Return to Command Mode	Esc or Ctrl + [	Exit insert mode

4. SEARCH & REPLACE OPERATIONS

Search Operations

Task	Command	Description
Forward Search	/word + Enter	Search forward for "word"
Backward Search	?word + Enter	Search backward for "word"
Next Match	n	Go to next search result
Previous Match	N	Go to previous search result
Search Current Word	*	Search for word under cursor
Case Insensitive Search	:set ignorecase	Enable case-insensitive search

Replace Operations

Task	Command	Description
Replace All in File	:%s/old/new/g	Replace all "old" with "new"
Replace with Confirmation	:%s/old/new/gc	Replace with confirmation prompt
Replace in Current Line	:s/old/new/g	Replace in current line only
Replace in Visual Selection	Select text + :s/old/new/g	Replace in selected area
Case Insensitive Replace	:%s/old/new/gi	Replace ignoring case

5. SAVE & QUIT COMMANDS

File Operations

Purpose	Command	Description
Save File	<code>:w</code>	Write changes to file
Save As	<code>:w newfilename</code>	Save with different name
Quit	<code>:q</code>	Quit Vim
Save and Quit	<code>:wq</code> or <code>:x</code>	Save changes and exit
Force Quit	<code>:q!</code>	Quit without saving
Force Save & Quit	<code>:wq!</code>	Force save and quit
Save All & Quit	<code>:xa</code>	Save all files and exit

6. TERMINAL INTEGRATION

Paste Operations by Environment

Environment	Paste Method	Description
Linux Terminal	<code>Ctrl + Shift + V</code>	Standard Linux paste
PuTTY / Windows	Right-click	Default PuTTY behavior
macOS Terminal	<code>Cmd + V</code>	Standard macOS paste
Vim Paste Mode	<code>:set paste</code>	Enter paste mode (prevents formatting issues)
Exit Paste Mode	<code>:set nopaste</code>	Return to normal mode

Clipboard Integration

Action	Command	Description
Copy to System Clipboard	<code>" +y</code>	Yank to system clipboard
Paste from System Clipboard	<code>" +p</code>	Paste from system clipboard
Cut to System Clipboard	<code>" +d</code>	Delete to system clipboard

7. MEMORY HOOKS & PRO TIPS

Essential Mnemonics

- "Gg - go, yy - yank, dd - delete, :wq - done!"

- "i for Insert, Esc for Escape"
- "/" for Find, :%s/ for Replace"

Productivity Tips

```
# Open file at specific line
vim +25 filename.txt

# Open multiple files
vim file1.txt file2.txt
:n          # Next file
:N          # Previous file

# Split screen editing
:sp file2.txt    # Horizontal split
:vsp file2.txt   # Vertical split
Ctrl + w + w    # Switch between splits
```

Common Workflows

```
# Quick editing session
vim config.txt    # Open file
/text_to_find     # Search for text
i                # Edit text
Esc              # Return to command mode
:wq              # Save and exit

# Bulk operations
:10,20s/old/new/g # Replace between lines 10-20
:g/pattern/d      # Delete all lines containing pattern
```

LINUX USER & GROUP MANAGEMENT - COMPREHENSIVE CHEAT SHEET

1. VIEW EXISTING ACCOUNTS

User Information Commands

What You Want	Command	Description & Output
All Local Users	<code>cat /etc/passwd</code>	Raw user database file
All Local Users (System)	<code>getent passwd</code>	System database query
Specific User Info	<code>getent passwd username</code>	Single user details

What You Want	Command	Description & Output
Current User ID	<code>id</code>	Current user's UID, GID, and groups
Specific User ID	<code>id username</code>	User's UID, GID, and group memberships
Who is Logged In	<code>who</code> or <code>w</code>	Currently logged in users
Last Logins	<code>last</code>	Login history

Group Information Commands

What You Want	Command	Description
All Local Groups	<code>cat /etc/group</code>	Raw group database file
All Local Groups (System)	<code>getent group</code>	System database query
User's Group Memberships	<code>groups username</code>	Groups a user belongs to
Specific Group Info	<code>getent group groupname</code>	Single group details

Understanding /etc/passwd Format

```
username:x:UID:GID:Full Name:/home/username:/bin/bash
```

- **Username:** Login name
- **x:** Password placeholder (stored in /etc/shadow)
- **UID:** User ID (0=root, 1-999=system, 1000+=regular)
- **GID:** Primary Group ID
- **Full Name:** GECOS field (user info)
- **Home Directory:** User's home path
- **Shell:** Default shell

2. ADD / DELETE USERS 🧑💻

User Creation Commands

Task	Command	Options & Description
Create Basic User	<code>useradd username</code>	Minimal user creation
Create User with Home	<code>useradd -m username</code>	Create with home directory

Task	Command	Options & Description
Create with Specific UID	<code>useradd -u 1500 username</code>	Assign specific User ID
Create with Specific GID	<code>useradd -g 1001 username</code>	Assign specific Primary Group
Create with Comment	<code>useradd -c "Full Name" username</code>	Add user description
Create with Specific Shell	<code>useradd -s /bin/bash username</code>	Set default shell
Create with Home Path	<code>useradd -m -d /custom/home username</code>	Custom home directory

User Deletion Commands

Task	Command	Effect
Remove User Only	<code>userdel username</code>	Delete user account only
Remove User + Home	<code>userdel -r username</code>	Delete user and home directory
Force Removal	<code>userdel -f username</code>	Force delete logged-in user

Password Management

Task	Command	Description
Set User Password	<code>passwd username</code>	Set/change password for user
Change Own Password	<code>passwd</code>	Change current user's password
Lock User Account	<code>passwd -l username</code>	Lock user account
Unlock User Account	<code>passwd -u username</code>	Unlock user account
Expire Password	<code>passwd -e username</code>	Force password change on next login

Key Facts About User Creation

- Automatic Primary Group:** Created with same name as user
- Home Directory:** Typically `/home/username` (with `-m` flag)
- Default Shell:** Usually `/bin/bash`
- System Users:** UID 1-999 (varies by distribution)
- Regular Users:** UID 1000+ (varies by distribution)

3. SWITCH USERS & SESSIONS

User Switching Commands

Task	Command	Description
Switch to User	<code>su - username</code>	Login as user with full environment
Switch to User (minimal)	<code>su username</code>	Switch without full environment
Become Root	<code>su -</code> or <code>sudo -i</code>	Switch to root user
Return to Previous	<code>exit</code> or <code>Ctrl + D</code>	Exit current shell
Run Single Command as User	<code>sudo -u username command</code>	Execute command as specific user

Session Management

```
# Multiple user switching example
whoami           # Check current user
su - john        # Switch to john
exit             # Return to original user
sudo -u john whoami # Run command as john without switching
```

4. GROUP OPERATIONS

Group Management Commands

Need	Command	Description
Create Group	<code>groupadd groupname</code>	Create new group
Create System Group	<code>groupadd -r groupname</code>	Create system group (GID < 1000)
Create with Specific GID	<code>groupadd -g 1500 groupname</code>	Create group with specific GID
Delete Group	<code>groupdel groupname</code>	Remove group
Modify Group	<code>groupmod -n newname oldname</code>	Rename group

User Group Membership

Need	Command	Description
Add User to Secondary Group	<code>usermod -aG groupname username</code>	Append user to group
Set User Primary Group	<code>usermod -g groupname username</code>	Change user's primary group
Add Multiple Groups	<code>usermod -aG group1,group2 username</code>	Add to multiple groups

Need	Command	Description
Remove from Group	<code>gpasswd -d username groupname</code>	Remove user from group
View User Groups	<code>groups username</code>	Show user's group membership

Group Password & Administration

Task	Command	Description
Set Group Password	<code>gpasswd groupname</code>	Set group password
Add Group Admin	<code>gpasswd -A username groupname</code>	Add group administrator
Remove Group Admin	<code>gpasswd -A "" groupname</code>	Remove all administrators

5. HOME DIRECTORY MANAGEMENT

Home Directory Operations

Task	Command	Description
List All Home Directories	<code>ls -la /home/</code>	View all user homes
Check Specific Home	<code>ls -la /home/username/</code>	Check user's home contents
Create Home Directory	<code>mkhomedir_helper username</code>	Create home directory for existing user
Set Home Directory	<code>usermod -d /new/home username</code>	Change user's home directory
Move Home Directory	<code>usermod -m -d /new/home username</code>	Move content to new home

Permission & Security

```
# Check home directory permissions
ls -ld /home/username/

# Typical secure permissions:
# drwx----- username username /home/username
# (Only owner can read/write/execute)

# Fix insecure permissions
chmod 700 /home/username/
chown username:username /home/username/
```

6. PRACTICAL SCENARIOS & EXAMPLES

Create Development Team

```
# Create groups
groupadd developers
groupadd designers

# Create users with home directories
useradd -m -c "John Developer" -G developers johndev
useradd -m -c "Jane Designer" -G designers janedesign

# Set passwords
passwd johndev
passwd janedesign

# Add user to multiple groups
usermod -aG developers,designers adminuser
```

User Account Maintenance

```
# Check account status
chage -l username    # Password aging info
last username        # Login history

# Lock and unlock accounts
usermod -L username  # Lock account
usermod -U username  # Unlock account

# Set account expiration
usermod -e 2024-12-31 username  # Expire account on date
```

Batch User Operations

```
# Create multiple users
for user in alice bob charlie; do
    useradd -m $user
    echo "$user:password123" | chpasswd
done

# Delete multiple users
for user in alice bob charlie; do
    userdel -r $user
done
```

7. SECURITY BEST PRACTICES

Account Security

- Use `passwd -l` to lock inactive accounts
- Regularly review `/etc/passwd` for unauthorized users
- Monitor `/var/log/secure` for authentication attempts

- Use `chage` to enforce password policies

Permission Guidelines

- Home directories should be 700 (drwx-----)
- Users should not have write access to other home directories
- Use groups for shared resource access instead of loose permissions